# Maxwell Higdon

Mechanical Engineer | Hardware Design and Systems Integration

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## PROFESSIONAL SUMMARY

Mechanical Engineer with 3+ years of experience in mechanical design, hardware development, and production-scale electromechanical systems. Proficient in SolidWorks, engineering drawings, mechanical troubleshooting, and developing design improvements for real-world applications. Experienced carrying engineering concepts and field-driven modifications through design documentation, fabrication, and implementation. Enjoys hands-on problem-solving, collaborative design, and developing practical mechanical solutions.

## TECHNICAL SKILLS

- Design & CAD: SolidWorks (CSWA), AutoCAD, Revit, Trace3D; GD&T; custom fabrication drawings
- Analysis: Thermal systems, fluid systems & flow distribution, predictive heat exchange modeling, root cause analysis
- Systems: Electromechanical integration, fluid transport, HVAC & plumbing integration, system troubleshooting
- Field & Process: Global installation support, site-specific modifications, cross-functional technical leadership

## PROFESSIONAL EXPERIENCE

Senior Engineer, Hardware | Ricoh USA

*Boulder, CO | April 2025 – Present*

- Led cross-functional teams in designing site-specific modifications for Ricoh Pro VC80000 inkjet press, improving reliability and deployment efficiency.
- Designed and implemented upgraded cooling fluid transport systems, optimizing flow distribution and thermal performance for over 15+ field production systems.
- Diagnosed and resolved complex electromechanical failures in live production environments, minimizing customer downtime.

Associate Engineer, Hardware | Ricoh USA

*Boulder, CO | July 2023 – April 2025*

- Provided technical support for global installations of continuous form inkjet platforms at customer sites.
- Revised engineering drawings to support custom component fabrication and field-driven design changes.
- Worked with senior engineers to troubleshoot system-level issues across mechanical, fluid, and control systems.

Student Intern II, MEP | Olsson

*Denver, CO | January 2023 – May 2023*

- Modeled HVAC and plumbing systems in Revit and AutoCAD for hospitals and commercial buildings to meet project and code requirements.
- Developed internal reference standards that improved consistency and efficiency across engineering workflows.
- Supported system layout design and coordination with multidisciplinary teams for construction documentation.

## PROJECTS

- Wheelchair-Compatible Motorized Disc Launcher | QL Plus | 2022 – 2023
  - Engineered and evaluated a functional wheelchair-accessible motorized disc launcher with an engineering team from initial concept to physical prototype.
- Assistive Audio Regulation Tool | University of Colorado Boulder | 2019 – 2020
  - Co-designed, fabricated, and validated a working prototype as part of a multidisciplinary team.
- Geothermal Pile Heat Exchanger | Cole Garner Geotechnical | 2021
  - Developed a predictive tool to estimate temperature change and heat exchange performance.

## EDUCATION, CERTIFICATIONS & AWARDS

- B.S., Mechanical Engineering, University of Colorado Boulder | 2019 – 2023
- Certified SolidWorks Associate (CSWA), Dassault Systèmes | 2020
- Recognition Award for Onsite Support Excellence, Ricoh USA | 2025
- Recognition Award for Business Development, Ricoh USA | 2025